

Auctions as Mechanism Design

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What is a mechanism?

- A mechanism is a process that a seller uses in allocating an object to potential buyers.
- We consider privately informed buyers.
- A mechanism has 3 components:
 - B - a set of possible bids (messages), one for each buyer (ex: $(b_1, b_2, \dots, b_N) \in B$);
 - $\pi : B \rightarrow \Delta$ - an allocation rule that maps bids into a probability distribution over N (i.e. $\pi_i(b)$ is a probability of the object being allocated to i);
 - $M : B \rightarrow \mathbf{R}^N$ - payment rule that maps bids into payments (i.e. $\mu_i(b)$ is the payment by buyer i);

Examples

- FPA(0,0) is a mechanism with $b_i \in [0, \infty)$ and

$$\pi_i^I(b_i) = \begin{cases} 0 & \text{if } b_i < \max_{j \neq i} b_j \\ 1 & \text{if } b_i > \max_{j \neq i} b_j \end{cases}$$

$$\mu_i^I(b_i) = \begin{cases} 0 & \text{if } b_i < \max_{j \neq i} b_j \\ b_i & \text{if } b_i > \max_{j \neq i} b_j \end{cases}$$

- SPA(0,0) differs from FPA(0,0) only in the payment rule:

$$\mu_i^{II}(b_i) = \begin{cases} 0 & \text{if } b_i < \max_{j \neq i} b_j \\ \max_{j \neq i} b_j & \text{if } b_i > \max_{j \neq i} b_j \end{cases}$$

Direct Mechanism and Revelation Principle

- Suppose now that instead of sending a message b_i , the mechanism requires $B = \tilde{V}$ where $\tilde{V} = (\tilde{v}_1, \dots, \tilde{v}_N)$.
- Then, the pair $(Q(\tilde{v}), P(\tilde{v}))$ denotes the outcome of the mechanism specifying the allocation rule $Q(\tilde{v})$ and the payment rule $P(\tilde{v})$ based on the valuation reports submitted by each bidder.

Revelation Principle

Given a mechanism (B, π, M) and a BNE for this mechanism, there exists a direct mechanism $(\tilde{V}, Q, P)$ in which:

- it is an equilibrium for each buyer to report his/her true value;
- the outcomes are the same as in the given equilibrium of the original mechanism.

Revelation Principle

- In order for the BNE to be implemented as truthful reporting, it is sufficient to set $Q(\tilde{v}) = \pi(b(\tilde{v}))$ and $P(\tilde{v}) = M(b(\tilde{v}))$.
- Thus, the direct mechanism ask the buyers to report their valuations and makes the “equilibrium calculations” for them.
- $(q_i(\tilde{v}_i, v_{-i}), p_i(\tilde{v}_i, v_{-i}))$ denote the probability of winning for buyer i and the payment given the set of reports $(\tilde{v}_i, v_{-i})$.
- Then from i 's perspective, the expected probability of winning is

$$Q_i(\tilde{v}_i) = \int_{[0,1]^{n-1}} q_i(\tilde{v}_i, v_{-i}) \Pi_{j \neq i} f(v_j) dv_j$$

- Similarly, from i 's perspective, the expected payment is

$$P_i(\tilde{v}_i) = \int_{[0,1]^{n-1}} p_i(\tilde{v}_i, v_{-i}) \Pi_{j \neq i} f(v_j) dv_j$$

Optimization Problem

- Buyer i 's interim expected utility from reporting $\tilde{v}_i$ when all others are reporting truthfully is:

$$Q_i(\tilde{v}_i)v_i - P_i(\tilde{v}_i)$$

- Maximizing the above expression with respect to $\tilde{v}_i$ gives the following FOC:

$$FOC : Q'_i(\tilde{v}_i)v_i - P'_i(\tilde{v}_i) = 0$$

Incentive Compatibility

- Incentive Compatibility requires that the solution of the above equation is v_i :

$$IC : Q'_i(v_i)v_i - P'_i(v_i) = 0 \quad (1)$$

- Second-order conditions are satisfied if:

$$Q''_i(v_i)v_i - P''_i(v_i) \leq 0 \quad (2)$$

IC constraint and properties of BNE

- Let $\Pi_i(v_i)$ denote the expected equilibrium payoff in the direct allocation mechanism.

Proposition 1

The IC constraints given by (1) and (2) are satisfied if and only if:

$$\Pi_i(v_i) = \Pi_i(0) + \int_0^{v_i} Q_i(y) dy \quad (3)$$

$$Q'_i(v_i) \geq 0 \quad (4)$$

- Equation (3) states that the surplus of bidder i in a BNE of any allocation mechanism depends only on his equilibrium probability of winning and the expected payoff received by the lowest type and not the payment rule.
- Equation (4) states that the probability that bidder i gets the object is non-decreasing in his valuation.

Proof

$$\Pi_i(v_i) = Q_i(v_i)v_i - P_i(v_i)$$

Differentiating w.r.t. v_i obtains:

$$\Pi'_i(v_i) = Q_i(v_i) + (Q'_i(v_i)v_i - P'_i(v_i)) = Q_i(v_i)$$

Integrating both sides obtains equation (3). To obtain equation (4), differentiate the IC constraint given by (1):

$$Q'_i(v_i) + (Q''_i(v_i)v_i - P''_i(v_i)) = 0$$

Since the second term is non-positive due to SOC, the first term needs to be non-negative.

Derivation of BNE of an auction

- Proposition 1 can be used to derive the bidding function of a specific auction.
- Consider FPA(r, c), in which all types above some v_0 participate and use a strictly increasing bid function.
- The expected probability of winning in this case is:

$$Q_i(v) = \begin{cases} 0 & \text{if } v < v_0 \\ G(v) & \text{if } v > v_0 \end{cases}$$

- $\Pi_i(v) = 0$ for all $v < v_0$ since they do not participate.

- The equilibrium payoff in BNE is:

$$\Pi_i(v) = G(v)v - P_i(v) = \int_{v_0}^v G(y)dy$$

- The expected payment:

$$P_i(v) = c + G(v)b_i(v)$$

Derivation of BNE of an auction

- Solving for $b_i(v)$ from the above equations obtains:

$$b_i(v) = v - \int_{v_0}^v \frac{G(y)}{G(v)} dy - \frac{c}{G(v)}$$

- The marginal guy v_0 wins only if all other bidders choose to stay out. Then it does not pay to bid more than r .

$$(v_0 - r)G(v_0) - c = 0$$

- $b(v)$ can be rewritten as:

$$b(v) = \int_{v_0}^v y \frac{g(y)}{G(y)} dy + r \frac{G(v_0)}{G(v)}$$

Equivalent Auctions

- If two auctions A and B are such that:
 - $\Pi_i^A(0) = \Pi_i^B(0)$
 - $Q_i^A(v_i) = Q_i^B(v_i)$
 - then $\Pi_i^A = \Pi_i^B$ and the two auctions are equivalent from the bidder i 's point of view.
- First and second price auctions are equivalent since:
 - $v_0(c, r)$ is the same for both auctions;
 - $\Pi_i(v) = 0$ for $v_i < v_0(c, r)$;
 - $Q_i(v_i) = G(v_i)$ for $v_i > v_0(c, r)$ and $Q_i(v_i) = 0$ otherwise.

Seller's Revenue Equivalence

Proposition 2

The seller's expected payoff in any BNE of any auction is:

$$\Pi_s(v_s) = v_s + \sum_{i=1}^N \left[\int_0^1 \left(v_i - \frac{1 - F(v_i)}{f(v_i)} - v_s \right) Q_i(v_i) f(v_i) dv_i - \Pi_i(0) \right] \quad (5)$$

Seller's Revenue Equivalence

Optimal Auction

- The seller would like to choose $Q_i(v_i)$ and $\Pi_i(0)$ to maximize equation (5) subject to IC constraints given by Proposition 1.

$$\Pi_s(v_s) = v_s + \sum_{i=1}^N \left[\int_0^1 \left(v_i - \frac{1 - F(v_i)}{f(v_i)} - v_s \right) Q_i(v_i) f(v_i) dv_i - \Pi_i(0) \right]$$

- Set $\Pi_i(0) = 0$
- Set $Q_i(v_i)$ to maximize:

$$\sum_{i=1}^N \int_0^1 \left(v_i - \frac{1 - F(v_i)}{f(v_i)} - v_s \right) Q_i(v_i) f(v_i) dv_i$$

$$s.t. \ Q'_i(v_i) \geq 0$$

Optimal Auction

- Let $\mathbf{v} = (v_1, \dots, v_N)$. Recall that

$$Q_i(v_i) = \int_{[0,1]^{n-1}} q_i(\mathbf{v}) \Pi_{j \neq i} f(v_j) dv_j$$

- Substituting for $Q_i(v_i)$ and moving the summation inside the integral, we obtain:

$$\int_{[0,1]^n} \left(\sum_{i=1}^N \left(v_i - \frac{1 - F(v_i)}{f(v_i)} - v_s \right) q_i(\mathbf{v}) \right) \Pi_{j=1}^N f(v_j) dv_j$$

$$s.t. \ q_i(\mathbf{v}) \geq 0 \ \forall i$$

$$\sum_{i=1}^N q_i(\mathbf{v}) \leq 1$$

$$q_i(\mathbf{v}) \text{ non-decreasing in } v_i$$

Optimal Auction

- This boils down to maximizing the summation s.t. the constraints.

$$\sum_{i=1}^N \left(v_i - \frac{1 - F(v_i)}{f(v_i)} - v_s \right) q_i(\mathbf{v})$$

- Let $v_i - \frac{1 - F(v_i)}{f(v_i)}$ be increasing in v_i . Then,

$$v_0^* - \frac{1 - F(v_0^*)}{f(v_0^*)} = v_s$$

- Optimality requires:

- $q_i^*(\mathbf{v}) = 1$ if $v_i = \max\{v_1, \dots, v_N\}$ and $v_i \geq v_0^*$
- $q_i^*(\mathbf{v}) = 0$ if $v_i \neq \max\{v_1, \dots, v_N\}$.

The Optimal Auction Theorem

Proposition 3

If $v - \frac{1-F(v)}{f(v)}$ is monotone increasing, then

FPA(r,c), SPA(r,c), EA(r,c), DA(r,c) is an optimal sales mechanism for the seller so long as the reserve price and/or entry fee is set so that

$$v_0 = v_0^*.$$

- The Optimal Auction is not necessarily ex-post efficient? Why?

References

- Myerson R. "Optimal Auction Design", Mathematics of Operations Research, 1981
- V.J. Krishna, "Auction Theory"- Chapter 5